*Last updated in October 2026*

# Yuri Ishitoya

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## RESEARCH INTERESTS

Multimodal foundation models for embodied AI, especially **low-cost ways to add or strengthen capabilities of pretrained models while preserving what they already know**, from VLM-to-VLA adaptation to interventions on internal representations.

## EDUCATION

**Rice University**, *Ph.D. in Computer Science* (Advisor: Prof. Vicente Ordóñez Román) — Houston, TX, USA
Toyota Riken Overseas Scholar — Aug 2026 – Present

**Ochanomizu University**, *M.S. Program in Information Science* (Advisor: Prof. Ichiro Kobayashi) — Tokyo, JPN
GPA: 4.0/4.0. Completed coursework; withdrew to begin Ph.D. at Rice (degree not conferred). — Apr 2025 – Aug 2026

**Ochanomizu University**, *B.S. in Information Science* — Tokyo, JPN
GPA: 3.8/4.0 — Apr 2021 – Mar 2025

## RESEARCH EXPERIENCE

**OMRON SINIC X**, *Research Intern* — Tokyo, JPN
Host: Dr. Mai Nishimura — Jan 2026 – Jul 2026

- Developed **CLAP** (project page), which converts a pretrained VLM (Qwen3.5) into a VLA policy with **zero architectural changes** by grounding numeric action tokens in natural-language action descriptions, keeping fine-tuning close to the model's pretrained distribution.
- **90.8% success on LIBERO** after only 8 epochs (2B), **+14.9 pt** over VLA-0 under the same backbone and budget; gains hold across 0.8B–2B scales, with improved robustness on LIBERO-PRO under language, object, and spatial perturbations.
- Deployed the 2B model on a **UR5e** arm: **60% real-robot pick-and-place success**, maintained even with **out-of-distribution distractors**. Releasing an open-weight model family (0.8B–4B).
- Co-first author; paper **accepted to CoRL 2026** [1].

**Rice University**, *Ph.D. Researcher* — Houston, TX, USA
Vision & Language Lab (Advisor: Prof. Vicente Ordóñez Román) — Aug 2026 – Present

- Investigating whether intervening on multimodal models' internal states, using visual information as a language-agnostic anchor, can lift low-resource-language performance toward high-resource levels without retraining (ongoing).

**Rice University**, *Visiting Researcher* — Houston, TX, USA
Vision & Language Lab (Host: Prof. Vicente Ordóñez Román) — Feb 2025 – Aug 2025

- Co-first author [2]: designed evaluation protocols (classification, bias, identity recognition) for a people-centric benchmark and evaluated **39 VLMs**; found SigLIP labels ~40–50% of female politicians as "teacher" (vs. ~5% of male), and a 72B generative VLM recognizes far fewer U.S. Senators than CLIP ViT-L/14 (45.8% vs. 94.9%).

**Ochanomizu University**, *Graduate & Undergraduate Researcher* — Tokyo, JPN
Kobayashi Lab, NLP (Advisor: Prof. Ichiro Kobayashi) — Apr 2024 – Aug 2026

- Proposed a dual intervention on LLM output embeddings (channel steering + gender-subspace projection) that mitigates both probabilistic and semantic bias, outperforming steering-vector baselines [3].
- Proposed a frequency-domain (DFT) analysis of log-likelihood sequences that separates stable bias signals from syntactic noise (B.S. thesis) [6].

**Fujitsu Ltd.**, *Research Intern* — Tokyo, JPN
Artificial Intelligence Research Center (Host: Dr. Hiroya Inakoshi) — Aug 2024

- Built an LLM-based synthetic dialogue pipeline with persona conditioning for training bias detectors (**86% accuracy**) [4, 5].

## PUBLICATIONS

***Refereed Conference Papers***

1. **Ishitoya, Yuri**[*], Siburian, J.[*], Hamaya, M., Saito, K., Beltran-Hernandez, C. C. & Nishimura, M.[*]. *CLAP: Direct VLM-to-VLA Adaptation via Language-Action Grounding* in *Proceedings of the Conference on Robot Learning (CoRL)* (2026). arXiv: 2607.08974. [*]Equal contribution.

***Manuscripts in Preparation***

2. **Ishitoya, Yuri**[*], Flores, V.[*], Yang, Z., Cascante-Bonilla, P. & Ordonez, V. *A People-centric Benchmark for Testing Vision-Language Models* 2025. [*]Equal contribution.

***Domestic Workshop & Conference Papers (non-refereed, in Japanese)***

3. **Ishitoya, Yuri** & Kobayashi, I. *Mitigating Statistical and Causal Bias Through Output Embedding Operations* in *Proceedings of the 32nd Annual Meeting of the Association for Natural Language Processing (NLP2026)* (2026).
4. Ishida, K., **Ishitoya, Yuri**, Kobayashi, I. & Igarashi, Y. *A Study on the Usefulness of an Interviewer Support System that Points Out Gender-Based Judgments* in *Proceedings of the 212th Human-Computer Interaction Study Group (IPSJ SIG-HCI)* (2025).
5. Ishida, K., **Ishitoya, Yuri**, Kobayashi, I. & Igarashi, Y. *Proposal and Development of an AI Interviewer Support System that Points Out Gender-Based Judgments* in *Proceedings of the 39th Annual Conference of the Japanese Society for Artificial Intelligence (JSAI2025)* (2025).
6. **Ishitoya, Yuri** & Kobayashi, I. *A Study on Detecting Subtle Biases by Expanding the Log-Likelihood into a Frequency Space* in *Proceedings of the 39th Annual Conference of the Japanese Society for Artificial Intelligence (JSAI2025)* (2025).

## FELLOWSHIPS & AWARDS

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| **Toyota Riken Overseas Scholarship**: full Ph.D. funding, USD 160,000 over two years | 2026 |
| **Tobitate! Study Abroad Scholarship**, MEXT: JPY 670K grant (22% acceptance) | 2025 |
| **Oin-kai Research Encouragement Award**, Ochanomizu University: JPY 100K prize | 2025 |
| **MITOU IT Program**, IPA, Government of Japan: JPY 2.74M grant (13% acceptance) | 2023 |
| **Women Leaders Support Fund**, Public Resource Foundation: JPY 1.0M grant (6.8% acceptance) | 2023 |

## TECHNICAL SKILLS

- **Programming:** Python, C, CUDA, SQL
- **Frameworks & tools:** PyTorch, Hugging Face Transformers, FSDP, Accelerate, NNsight, Weights & Biases, Docker
- **Methods:** VLM/VLA fine-tuning, RL (PPO, GRPO), activation steering & interpretability, sim & real-robot evaluation

## LEADERSHIP & SERVICE

- Founder, Dots to Code Association (2024–25): organized Japan's largest hackathon for women and non-binary engineers (**300+ participants**).
- Teaching Assistant, *Introduction to Natural Language Theory*, Ochanomizu University (Spring 2025).